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PHYSICS

1) A particle excutes SHM on a straight line path. The amplitude of oscillation is 2 cm. When the displacement of the particle from the mean position is 1 cm, the numerical value of magnitude of acceleration, is equal to the numerical value of magnitude of velocity. The frequency of SHM (in second^{-1}) is :-

(1) $2\pi\sqrt{3}$

(2) $\frac{2\pi}{\sqrt{3}}$

(3) $\frac{\sqrt{3}}{2\pi}$

(4) $\frac{1}{2\pi\sqrt{3}}$

2) A spring elongated by length '□' when a mass 'M' is suspended to it. Now a tiny mass 'm' is attached with it and then released, its time period of oscillation is :-

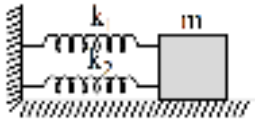
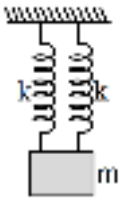
(1) $2\pi\sqrt{\frac{(M+m)\ell}{Mg}}$

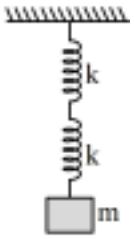
(2) $2\pi\sqrt{\frac{m\ell}{Mg}}$

(3) $2\pi\sqrt{\ell/g}$

(4) $2\pi\sqrt{\frac{M\ell}{(m+M)g}}$

3) Match the column-I with Column-II.

Column I		Column II	
(A)		(p)	$T = 2\pi\sqrt{\frac{m(k_1+k_2)}{k_1k_2}}$
(B)		(q)	$T = 2\pi\sqrt{\frac{2m}{k}}$

(C)		(r)	$T = 2\pi\sqrt{\frac{m}{2k}}$
(D)		(s)	$T = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$

(1) A-p, B-q, C-s, D-r

(2) A-s, B-r, C-p, D-q

(3) A-r, B-p, C-s, D-q

(4) A-p, B-r, C-q, D-s

4) A particle is moving on x-axis has potential energy U = 2 – 20 x + $5x^2$ joule. The particle is released at x = – 3. The maximum value of 'x' will be [x is in meter and U is in joule] :-

(1) 5 m

(2) 3 m

(3) 7 m

(4) 8 m

5) ***Assertion :*** Acceleration is proportional to the displacement; this condition is not sufficient for motion in simple harmonic.
Reason : In simple harmonic motion, direction of displacement is also considered.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

6) The time period of a simple pendulum is T. The relative density of bob is σ. The whole of the pendulum is immersed in water and allowed to oscillate. Assuming there are no viscous forces, the new time period of pendulum will be :–

(1) T(σ – 1)

(2) $T\sqrt{\frac{\sigma}{\sigma - 1}}$

(3) T/(σ – 1)

(4) $T\sqrt{\frac{\sigma - 1}{\sigma}}$

7) The potential energy of a particle of mass 1 kg in motion along the x-axis is given by U = 4(1 – cos2x), where x is in metres. The period of small oscillation (in second) is :-

(1) 2π

(2) π

(3) $\frac{\pi}{2}$

(4) $\sqrt{2}\pi$

8) A body is executing simple harmonic motion. At a displacement x, its potential energy is E_1 and at a displacement y, its potential energy is E_2. The potential energy at a displacement (x + y) is :

(1) $E_1 + E_2$

(2) $\sqrt{E_1^2 + E_2^2}$

(3) $E_1 + E_2 + 2\sqrt{E_1 E_2}$

(4) $\sqrt{E_1 E_2}$

9) If the amplitude of the SHM be A, then what is the displacement of the particle, when kinetic energy of SHM is 1/n times of its potential energy ?

(1) nA

(2) $\left(\sqrt{\frac{n+1}{n}}\right) A$

(3) $\left(\sqrt{\frac{n}{n+1}}\right) A$

(4) $\frac{A}{n}$

10) If the maximum velocity and maximum acceleration of a particle executing SHM are equal in magnitude, the time period will be :-

(1) 1.57 sec.

(2) 3.14 sec.

(3) 6.28 sec.

(4) 12.56 sec.

11) Four simple harmonic vibrations :

$y_1 = 8\cos(\omega t)$; $y_2 = 4\cos\left(\omega t + \frac{\pi}{2}\right)$

$y_3 = 2\cos(\omega t + \pi)$; $y_4 = \cos\left(\omega t + \frac{3\pi}{2}\right)$

are superimposed on each other. The resultant amplitude and phase are respectively :

(1) $\sqrt{45}$ and $\tan^{-1}\left(\frac{1}{2}\right)$

(2) $\sqrt{45}$ and $\tan^{-1}\left(\frac{1}{3}\right)$

(3) $\sqrt{75}$ and $\tan^{-1}\left(\frac{1}{2}\right)$

(4) $\sqrt{75}$ and $\tan^{-1}\left(\frac{1}{3}\right)$

12) Two vibrating strings A and B of same material but lengths 2L and 3L have radii 3r and 2r respectively. They are stretched under same tension. String A vibrates in fundamental mode and string B in second overtone. The ratio of their frequencies n_A/n_B will be :

(1) 1 : 2
(2) 1 : 3
(3) 1 : 4
(4) 2 : 3

13) Equation of a standing wave is generally expressed as y = 2A sin ωt coskx. In the equation, quantity ω/k represents :-

(1) the transverse speed of the particles of the string
(2) the speed of either of the component wave
(3) the speed of the standing wave
(4) a quantity that is independent of the properties of the string.

14) A wave represented by the equation y = a cos(kx – ωt) is superposed with another wave to form a stationary wave such that the point x = 0 is a node . The equation for other wave is :

(1) a sin (kx + ωt)
(2) –a cos(kx + ωt)
(3) –a cos(kx – ωt)
(4) –a sin(kx – ωt)

15) If a pipe gives notes of frequencies 255 Hz, 425 Hz and 595 Hz, what is fundamental frequency of the pipe and its type ?

(1) 85 Hz, open pipe
(2) 17 Hz, closed pipe
(3) 51 Hz, open pipe
(4) 85 Hz, closed pipe

16) In a gaseous medium, on increasing temperature by 800 K, speed of sound becomes $\sqrt{5}$ times of initial then initial temperature of medium in °C is:-

(1) 27°C
(2) 300°C
(3) –73°C
(4) 73°C

17) ***Assertion (A) :*** Due to end correction fundamental frequency of any pipe becomes less than frequency without end correction. ***Reason (R) :*** Due to end correction effective length of pipe increases.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

18) The equation of a simple harmonic wave is given by :-

$y = 3 \sin \frac{\pi}{2}(50\ t - x)$,

where x and y are in metres and t is in seconds. The ratio of maximum particle velocity to the wave velocity is :-

(1) 3π

(2) $\frac{2}{3\pi}$

(3) 2π

(4) $\frac{3}{2\pi}$

19) A transverse sinusoidal wave of amplitude a, wavelength λ and frequency f is travelling on a stretched string. The maximum speed of any point on the string is $\frac{v}{10}$, where v is the speed of propagation of the wave. If $a = 10^{-3}$ m and v = 10 m/s, then λ and f are given by :–

(I) $\lambda = 2\pi \times 10^{-2}$ m (II) $\lambda = 10^{-3}$ m

(III) $f = \frac{10^3}{2\pi}$ Hz (IV) $f = 10^4$ Hz

(1) I, III

(2) II, IV

(3) I, IV

(4) II, III

20) A wave equation which gives the displacement along Y-direction is given :-

$y = 10^4 \sin(60t + 2x)$

where x and y are in meter and t in sec. Among the following choose the correct statement :

(1) It represents a wave propagating along positive x-axis with a velocity of 30 m/s

(2) It represents a wave propagating along negative-x-axis with a velocity of 120 m/s

(3) It represents a wave propagating along negative x-axis with a velocity of 30 m/s

(4) It represents a wave propagating along negative x-axis with a velocity of 10^4 m/s

21) Transverse mechanical waves can be propagate in :–

(1) only liquid

(2) only gas

(3) only solid and only on liquid surface

(4) neither solid nor gas

22) Let speed of sound waves in hydrogen gas at room temperature is v_0. What will be the speed of sound waves in a room which contains an equimolar mixture of hydrogen and 'Helium' at same temperature :-

(1) $\sqrt{\frac{5}{7}}v_0$

(2) $\sqrt{\frac{7}{5}}v_0$

(3) $\sqrt{\frac{2}{5}}v_0$

(4) None of these

23) The lengths of two closed organ pipes are 0.750 m and 0.770 m. If they are sounded together, 3 beats per second are produced. The velocity of sound will be :-

(1) 350.5 m/sec

(2) 335.5 m/sec

(3) 346.5 m/sec

(4) None of these

24) The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to α times its original magnitude, where α equals :

(1) 0.7

(2) 0.81

(3) 0.729

(4) 0.6

25) In a resonance tube experiment, the first resonance is obtained for 10 cm length of air column and the second for 32 cm. The end correction for this apparatus is -

(1) 0.5 cm

(2) 1.0 cm

(3) 1.5 cm

(4) 2 cm

26) A block of mass 1.2 kg is heated to temperature of 1000°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx) (specific heat for the block = 0.1 cal/gm°C):-

(1) 1 kg

(2) 1.5kg

(3) 2 kg

(4) 4kg

27) Consider two containers A and B containing monoatomic gases at the same Pressure (P), Volume (V) and Temperature (T). The gas in A is compressed isothermally to $\frac{1}{8}$ of its original volume while

the gas B is compressed adiabatically to $\frac{1}{8}$ of its original volume. The ratio of final pressure of gas in B to that of gas in A is :

(1) 8

(2) $8^{\frac{3}{2}}$

(3) $\frac{1}{8}$

(4) 4

28) A diatomic gas is heated at constant pressure. The ratio dU : dQ : dW :

(1) 5 : 7 : 3

(2) 5 : 7 : 2

(3) 3 : 7 : 2

(4) 3 : 5 : 2

29) The equation of state of a gas is given by PV = nRT + Pb, here b = constant, n = number of moles. If 2 moles of a gas is isothermally expanded from volume V to 2V then work done during the process is

(1) $2RT\ln\left(\frac{2V-b}{V-b}\right)$

(2) $2RT\ln\left(\frac{2V}{V}\right)$

(3) $RT\ln\left(\frac{2V}{2V-b}\right)$

(4) $RT\ln\left(\frac{2V-b}{V-b}\right)$

30) In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation VT = K, where K is a constant. In this process the temperature of the gas is increased by ΔT. The amount of heat absorbed by gas is (R is gas constant) :

(1) $\frac{1}{2}R\Delta T$

(2) $\frac{3}{2}R\Delta T$

(3) $\frac{1}{2}KR\Delta T$

(4) $\frac{2K}{3}\Delta T$

31) A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °C, if this thermometer in the contact with the object reads $x_0/2$?

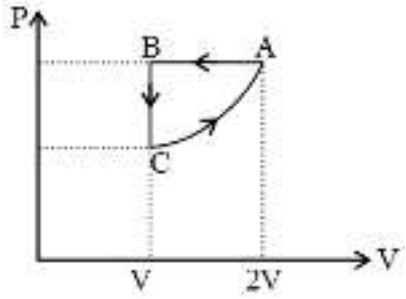
(1) 35

(2) 25

(3) 60

(4) 40

32) One mole of diatomic gas is taken through below cyclic process. The process CA is defined as P = (constant) V^2. Temperature at C is 300 K. Match the magnitudes of the quantities in Column I to those in Column II.



Column-I		Column-II	
(I)	Work done in path CA is	(P)	–1200 R
(II)	Work done in path AB is	(Q)	–2250 R
(III)	Change in internal energy in path BC is	(R)	700 R
(IV)	Heat transferred in path AB is	(S)	–4200 R

(1) (I) → P, (II) → R, (III) → Q, (IV) → S

(2) (I) → P, (II) → R, (III) → S, (IV) → Q

(3) (I) → R, (II) → P, (III) → S, (IV) → Q

(4) (I) → R, (II) → P, (III) → Q, (IV) → S

33) The power lost from a 20 cm thick slab of steel is 100 W. Assuming the temperature difference of 20 K, find the area of the slab.
(Thermal conductivity of steel = 45 W/mK)

(1) 0.011 m^2

(2) 0.022 m^2

(3) 0.033 m^2

(4) 0.044 m^2

34) A cylinder with movable piston contains 2 moles of hydrogen at standard temperature and pressure. The walls are made of heat insulator, piston is also well insulated. How much times the pressure of the gas becomes if gas is suddenly compressed to half of initial volume?
[Given $(2)^{1.4} = 2.64$]

(1) 1.5

(2) 2.64

(3) 2.00

(4) 4.00

35) One mole of an ideal monoatomic gas is compressed adiabatically from 6 Ltr to 2 Ltr. The initial pressure of the gas is 10^5 N/m^2. Then find final pressure of gas : [$(3)^{5/3} = 6.19$]

(1) 6.19×10^5 N/m^2

(2) $\frac{1}{6.19}$N/m^2

(3) 6.19 N/m^2

(4) $\frac{1}{6.19} \times 10^5$N/m^2

36) Radius of two spheres of same material are 1 & 4 m respectively and their temp, are 4×10^3 and 2×10^3 K respectively. Then ratio of emitted energy of spheres per sec. will be -

(1) 1 : 2

(2) 2 : 1

(3) 1 : 1

(4) 4 : 1

37) When the temperature of a rod increases from T to (T + ΔT), its moment of inertia increases from I to (I + ΔI). If α be the coefficient of linear expansion of the rod, then the value of $\left(\frac{\Delta I}{I}\right)$ is:

(1) 2α Δt

(2) α Δt

(3) $\frac{\alpha \Delta t}{2}$

(4) $\frac{\Delta t}{\alpha}$

38) On centigrade scale the temperature of a body increases by 30°. The increase in temperature on Fahrenheit scale is-

(1) 50°

(2) 40°

(3) 30°

(4) 54°

39) **Statement-I :-** First law of thermodynamics is based on energy conservation.
Statement-II :- Second law of thermodynamics put limitations on first law.

(1) Statement I and Statement II both are wrong.

(2) Statement I and Statement II both are right.

(3) Statement I is right but Statement II is wrong.

(4) Statement I is wrong but Statement II is right.

40) A carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased-

(1) 933 K

(2) 473 K

(3) 560 K

(4) 373 K

41) A calorimeter contains 0.2 kg of water at 30°C, 0.1kg of water at 60°C is added to it, the mixture is well stirred and the resulting temperature is found to be 35°C. The thermal capacity of calorimeter is :-

(1) 6300 J/K

(2) 1260 J/K

(3) 4200 J/K

(4) 8400 J/K

42) A solid sphere and a hollow sphere of same material and size are heated to same temperature and allowed to cool in the same surroundings. If the temperature difference between each sphere and its surroundings is T, then:

(1) the hollow sphere will cool at a faster rate for all values of T.

(2) the solid sphere will cool at a faster rate for all values of T

(3) both spheres will cool at the same rate for all values of T

(4) both spheres will cool at the same rate only for small values of T.

43)

If three molecules have velocities 1 km s^{-1}, 1 km s^{-1} and 1 km s^{-1}, the ratio of the rms speed and average speed is :-

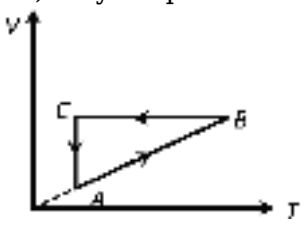
(1) 1

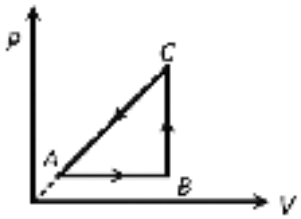
(2) 1.14

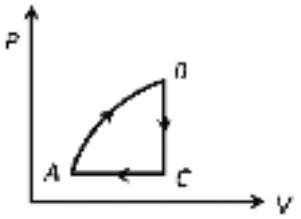
(3) 1.41

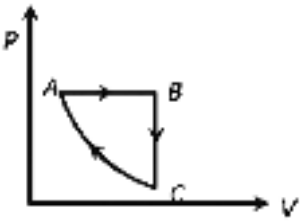
(4) 1.2

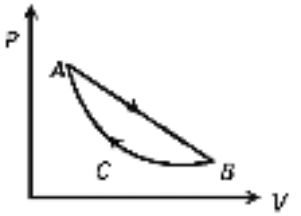
44) A cyclic process ABCA is shown in the V-T diagram. Process on the P-V diagram is :



(1) 

(2) 

(3) 

(4) 

45) The first resonance length of a resonance tube is 17 cm and the second resonance length is 53 cm. The third resonance length of the tube will be :-

(1) 85 cm

(2) 87 cm

(3) 88 cm

(4) 89 cm

CHEMISTRY

1) A steam volatile organic compound which is immiscible with water has a boiling point of 250°C. During steam distillation, a mixture of this organic compound and water will boil

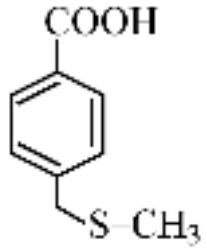
(1) above 100°C but below 250°C

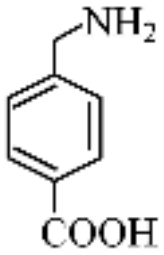
(2) above 250°C

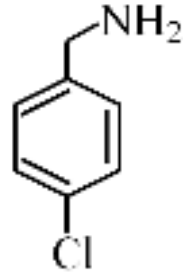
(3) at 250°C

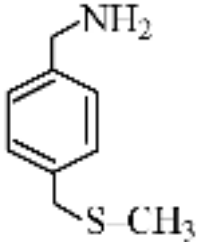
(4) close to but below 100°C

2) The compound that gives blood red colour in Lassaigne's test is :

(1) 

(2) 

(3) 

(4) 

3) Match list-I with list-II

	List-I		List-II
A.	Aniline + water	I.	Fractional distillation
B.	Aniline + Chloroform	II.	Simple distillation
C.	Crude oil	III.	Distillation under reduced pressure
D.	Separation of Glycerol from spentlye	IV.	Steam distillation

(1) A-(III), B-(IV), C-(II), D-(I)

(2) A-(IV), B-(I), C-(III), D-(II)

(3) A-(II), B-(IV), C-(I), D-(III)

(4) A-(IV), B-(II), C-(I), D-(III)

4) **Statement-I :-** In Dumas method the nitrogen containing organic compound is heated with copper oxide.

Statement-II :- In Kjeldahl's method, the nitrogen containing compound is heated with concentrated sulphuric acid.

(1) Statement-I & II both are correct

(2) Statement-I & II both are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

5)

In Carius method of estimation of halogen, 0.15 g of organic compound gives 0.12 g AgBr, then percentage of Bromine is:-
(At. wt of Br = 80, Molar mass of AgBr = 188)

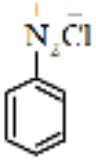
(1) 34%

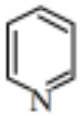
(2) 42%

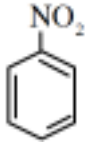
(3) 38%

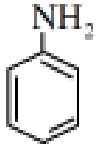
(4) 53%

6) Nitrogen can be estimated by Kjeldhal's method for which of the following compound ?

(1) 

(2) 

(3) 

(4) 

7) The number of σ bond and π bond in following compound respectively. $HO\text{-}CH(CN)_2$

(1) 6σ, 5π

(2) 6σ, 4π

(3) 7σ, 2π

(4) 7σ, 4π

8) Match List - I with List - II :-

	List-I (Common name)		**List-II (IUPAC name)**

(a)	Lactic acid	(i)	2-Methyl aniline
(b)	Mesityl oxide	(ii)	2-Hydroxy Propanoic acid
(c)	Acetanilide	(iii)	4-Methylpent-3-en-2-one
(d)	o-Toluidine	(iv)	N-Phenylethanamide

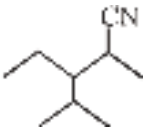
(1) a-(iii), b-(ii), c-(i), d-(iv)

(2) a-(iv), b-(ii), c-(i), d-(iii)

(3) a-(ii), b-(iii), c-(iv), d-(i)

(4) a-(ii), b-(iii), c-(i), d-(iv)

9)

The IUPAC name of  is :

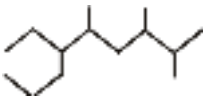
(1) 2-Cyano-3-ethyl-4-methylpentane

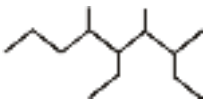
(2) 3-Isopropyl-2-methylpentanenitrile

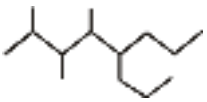
(3) 3-Ethyl-2, 4-dimethylpentanenitrile

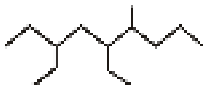
(4) 3-Isopropyl pentane-2-cyanide

10) The correct structure of
6-Ethyl-2,3,5-trimethyl nonane is :-

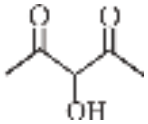
(1) 

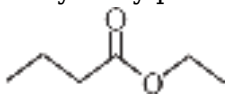
(2) 

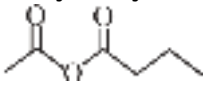
(3) 

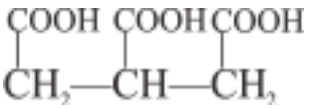
(4) 

11) Correct IUPAC match are :-

(a)  ;
3-Hydroxy pentane-2, 4-dione

(b)  ;
Ethyl butyrate

(c)  ;
Butanoic ethanoic anhydride

(d)  ;

Propane-1, 2, 3-tricarboxylic acid.

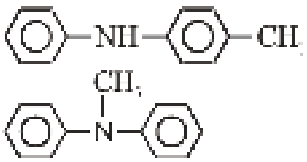
(1) a, b, c, d

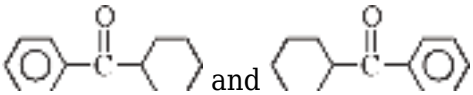
(2) a, c, d only

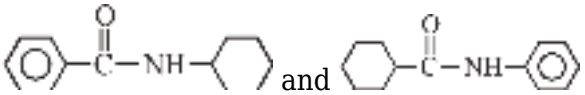
(3) a, d only

(4) b, c, d only

12) Which of the following pair are metamers ?

(1)  and

(2) 

(3) 

(4) All of these

13) The number of primary, secondary and tertiary structural alcohol possible with the formula $C_4H_{10}O$ is given by the set of :-

(1) 2,1,0

(2) 1,2,1

(3) 2,1,1

(4) 2,1,2

14)

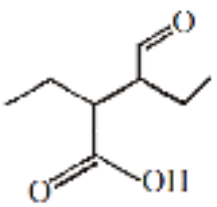
How many cyclic structural isomers form by C_5H_{10} :-

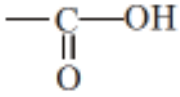
(1) 3

(2) 4

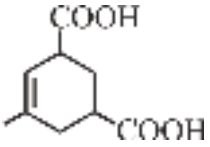
(3) 5

(4) 6

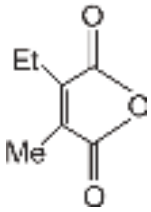
15) ***Assertion :-***  Principal carbon chain has 5 C-atoms in the given compound.

Reason :-  is the principal functional group of the compound.

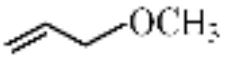
(1) Both Assertion and Reason are correct
(2) Assertion is correct but Reason is incorrect
(3) Assertion is incorrect but Reason is correct
(4) Both Assertion and Reason are incorrect

16) Correct IUPAC name of the following :- 

(1) 1-methylcyclohex-1-ene-3,5-dicarboxylic acid
(2) 5-methylcyclohex-4-ene-1,3-dicarboxylic acid
(3) 5-methylcyclohex-5-ene-1,3-dicarboxylic acid
(4) 5-methylcyclohex-4-ene-1,3-dioic acid

17) Correct IUPAC name of the compound is 

(1) 2-Ethyl-3-methylbut-2-ene-1, 4-dioic anhydride
(2) 3-Ethyl-2-methylbut-2-enedioic anhydride
(3) 2-Ethyl-3-Methyl-1,4-diketobut-2-enoic anhydride
(4) 2-Ethyl-3-methylcyclopentanoxy-1,4-dione

18) **Assertion :-**  and  are positional isomers of each other.
Reason :- Position of double bond change, so these are positional isomers.

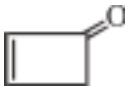
(1) Both Assertion and Reason are correct
(2) Assertion is correct but Reason is incorrect
(3) Assertion is incorrect but Reason is correct
(4) Both Assertion and Reason are incorrect

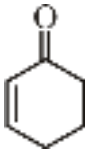
19) The correct IUPAC name of $[(CH_3)_3C]_4C$ is :-

(1) 3,3-sec-butyl-2,2,4,4-tetramethylpentane
(2) 3,3-ditertbutyl-2,2,4,4-tetramethylpentane
(3) Tetra t-butylmethane
(4) Tetra t-butylcarbon

20) Which of the following compound will have most stable enol ?

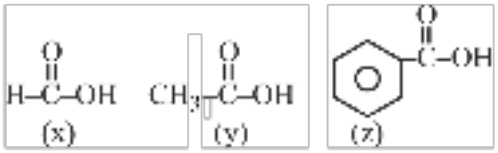
(1) 

(2) 

(3) 

(4) 

21) Arrange the following acids in decreasing order of acidity :-



(1) x > z > y

(2) x > y > z

(3) z > x > y

(4) z > y > x

22) How many of following are aromatic ?

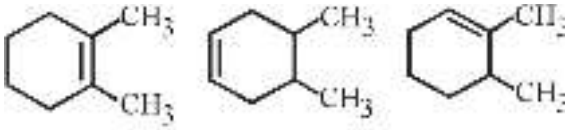


(1) 6

(2) 4

(3) 5

(4) 3



23) Which of following orders is correct for heat of hydrogenation of these compounds ?

(1) I > II > III

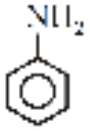
(2) III > II > I

(3) II > III > I

(4) III > I > II

24) Correct decreasing order of basic strength :-

(A) $(C_2H_5)_2NH$ (B) $C_2H_5NH_2$

(C) $(C_2H_5)_3N$ (D) 

(1) A > B > C > D

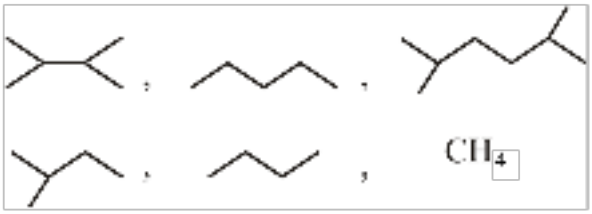
(2) D > A > C > B

(3) D > C > B > A

(4) A > C > B > D

25)

How many compound can be obtained in good yield by wurtz reaction.



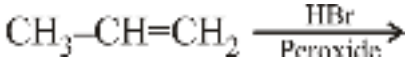
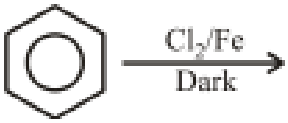
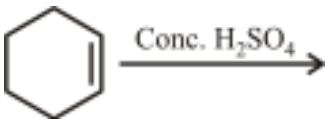
(1) 6

(2) 3

(3) 4

(4) 5

26) Match the reactions in column-I with their types in column-II

Column-I (Reaction)		**Column-II (Types)**	
(a)		(p)	Electrophilic addition reaction
(b)		(q)	Free radical addition reaction
(c)		(r)	Elimination reaction
(d)		(s)	Electrophilic substitution reaction

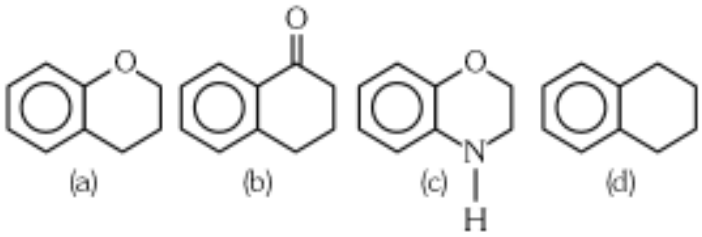
(1) a-r, b-q, c-s, d-p

(2) a-q, b-s, c-r, d-p

(3) a-p, b-r, c-s, d-q

(4) a-s, b-p, c-q, d-r

27) Rank the following compounds in decreasing order of reactivity towards electrophilic aromatic



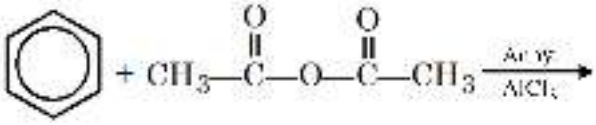
substitution reaction (ESR).

(1) (c) > (a) > (b) > (d)

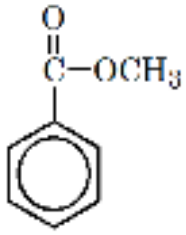
(2) (c) > (a) > (d) > (b)

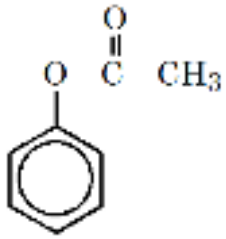
(3) (d) > (c) > (b) > (a)

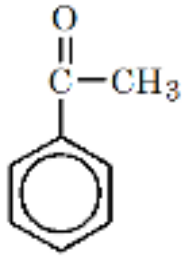
(4) (a) > (c) > (d) > (b)

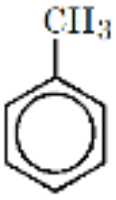
28) 

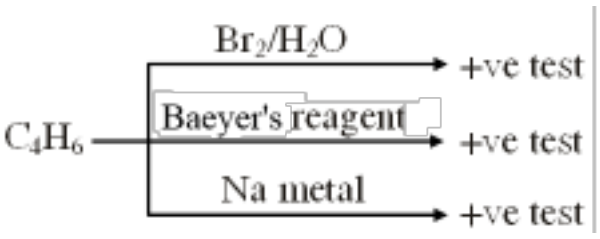
Final product of above reaction is :-

(1) 

(2) 

(3) 

(4) 

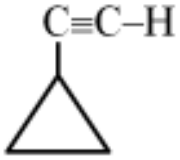
29) 

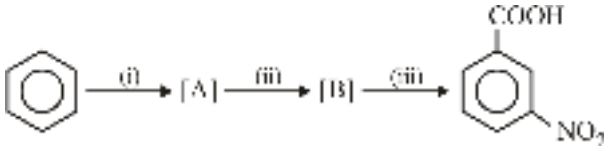
Compound X is :-

(1) $H_3C-C\equiv C-CH_3$

(2) $CH_2=CH-CH=CH_2$

(3) $CH_3-CH_2-C\equiv CH$

(4) 

30) 

reagents (i), (ii) & (iii) required for following conversion

(1) (i) H_3C–Cl/Anhydrous $AlCl_3$, (ii) $KMnO_4$, $H^{\oplus}$, (iii) Conc. HNO_3 + Conc. H_2SO_4

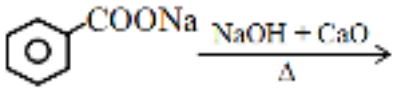
(2) (i) Conc. HNO_3 + Conc. H_2SO_4, (ii) $KMnO_4$, $H^{\oplus}$, (iii) H_3C–CH_2Cl/Anhydrous $AlCl_3$

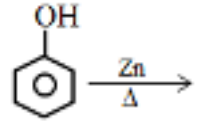
(3) (i) O_3/Zn-H_2O, (ii) $KMnO_4$, $H^{\oplus}$, (iii) Conc. NaOH

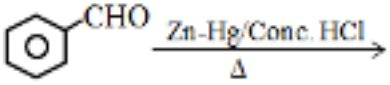
(4) (i) CO_2/NaOH, (ii) $CHCl_3$+NaOH, (iii) HNO_2

31) In which of the following reactions benzene is not formed in product?

(1) $HC\equiv CH \xrightarrow[873\ K]{\text{Red hot iron tube}}$

(2) 

(3) 

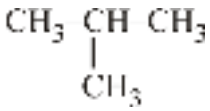
(4) 

32) $C_2H_5Cl \xrightarrow{Zn;H^{\oplus}}$ major product is :

(1) $CH_2=CH_2$

(2) CH_3–CH_2–CH_2–CH_3

(3) CH_3–CH_3

(4) 

33) The number of possible enantiomeric pairs that can be produced on Monochlorination of 2–methylbutane is :-

(1) 2

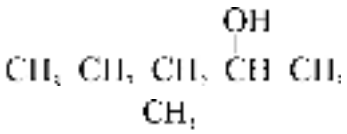
(2) 3

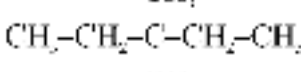
(3) 4

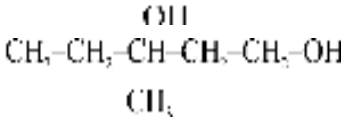
(4) 1

34) Among the following compounds which can be dehydrated most easily is -

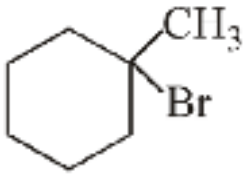
(1) $CH_3–CH_2–CH_2–CH_2–CH_2OH$

(2) 

(3) 

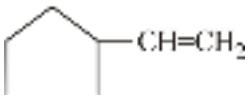
(4) 

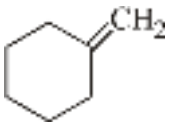
35) In the given reaction

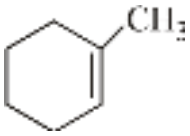


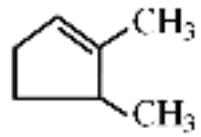
C_7H_{12} (X) $\xrightarrow{HBr}$ (Major product)

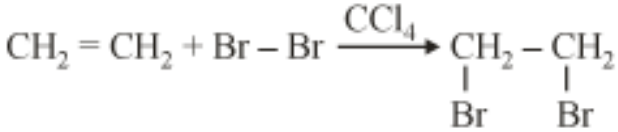
(X) can not be :

(1) 

(2) 

(3) 

(4) 



36) Ethene 1,2-dibromoethane

Correct set of statement of above reaction is -

(a) Red orange colour of bromine disappear after addition of bromine

(b) It is an example of electrophilic addition reaction.

(c) This reaction proceeds by formation of carbocation.

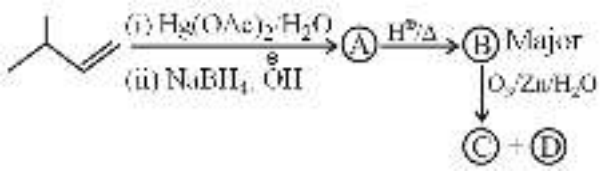
(d) Cyclic halonium ion is formed in this reaction.

(1) a, b, c only

(2) a, b, d only

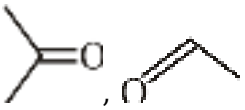
(3) a, d only

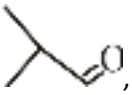
(4) a, b, c, d all

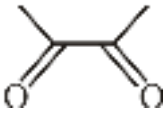


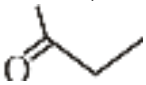
37)

C and D are :-

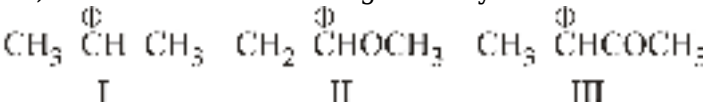
(1) 

(2)  , HCHO

(3) 
HCHO,

(4) 

38) The order of decreasing stability of the cations :-



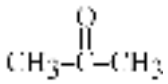
(1) III > II > I

(2) I > II > III

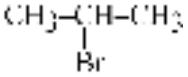
(3) II > I > III

(4) I > III > II

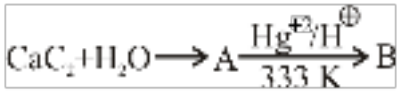
39) Electromeric effect is not observed in :

(1) 

(2) CH_3-C≡CH

(3) 

(4) $CH_3CH=CH-CH_3$

40) 

Identify A & B in the given reaction :-

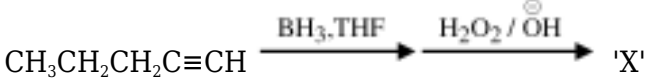
(1) C_2H_4 and HCOOH

(2) C_2H_4 and CH_3COOH

(3) C_2H_2 and CH_3CHO

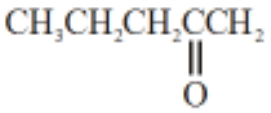
(4) C_2H_2 and CH_3COOH

41)

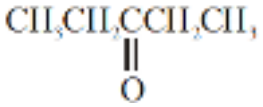


Product 'X' is :

(1) $CH_3CH_2CH_2CH_2CHO$

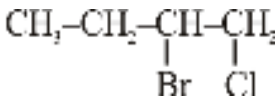
(2) 

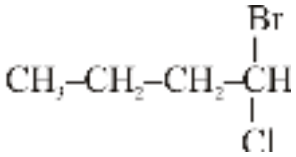
(3) $CH_3CH_2CH_2CHO$

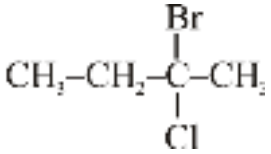
(4) 

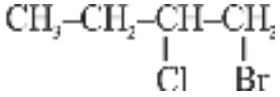
42) $CH_3–CH_2–C\equiv CH \xrightarrow{HCl} A \xrightarrow{HBr} B$

B is :-

(1) 

(2) 

(3) 

(4) 

43) Which of the following compound can not be used as halide component in friedel craft reaction?

(1) Bromomethane

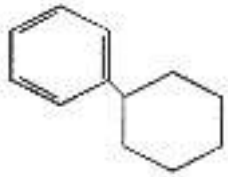
(2) Chloroethene

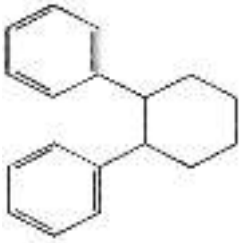
(3) 3-bromopropene

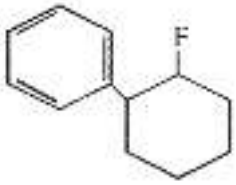
(4) Benzyl chloride

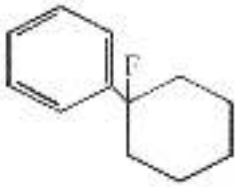
44) 

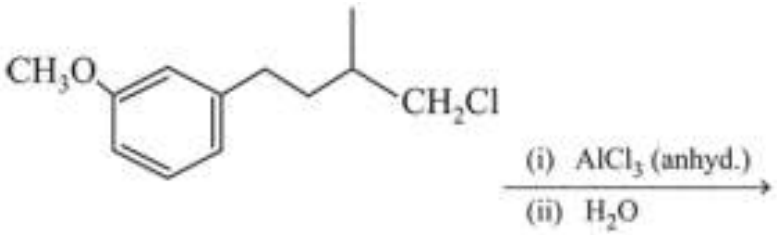
, the product P is :-

(1) 

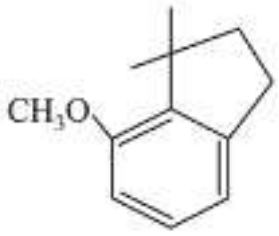
(2) 

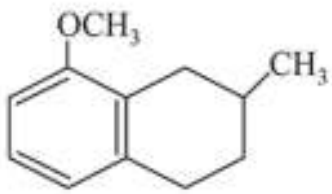
(3) 

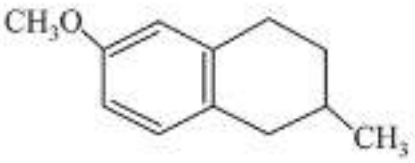
(4) 

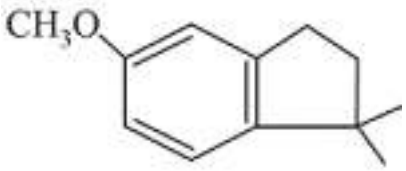


45)
'A' (Major product)

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Jan Ingenhousz experiment shows that :-

(1) CO_2 is essential for photosynthesis
(2) Sunlight is essential to the plant process that purifies the air
(3) Water is essential for photosynthesis
(4) Chlorophyll is essential for photosynthesis

2) First action spectrum was obtained by using :-

(1) Green algae
(2) Brown algae
(3) Red algae
(4) Blue green algae

3) Photochemical phase of photosynthesis include light absorption, water splitting, _____ and the formation of high-energy chemical intermediates, ATP and NADPH.

(1) Oxygen release
(2) Oxygen use
(3) Carbondioxide release
(4) Carbondioxide use

4) Which of the following required for chemiosmosis ?

(1) Proton pump
(2) ATP synthase
(3) A membrane
(4) All

5) What goes in the calvin cycle to make one molecule of glucose.

(1) $6CO_2$, 18 ATP, 12 NADPH

(2) 1 glucose, 18 ADP, 12 NADP

(3) 6 CO_2, 12 ATP, 18 NADPH

(4) 1 CO_2, 3ATP, 2 NADPH

6) Which of the following is first stable product in C_4 cycle ?

(1) Oxaloacetic acid

(2) RuBP

(3) Phosphoenol pyruvic acid

(4) 3-phosphoglyceric acid

7) How many CO_2 molecules are fixed in three cycles of C_2 cycle ?

(1) Three

(2) Six

(3) Eighteen

(4) Zero

8) Which of the following is major limiting factor for photosynthesis?

(1) Light

(2) CO_2

(3) Temperature

(4) Water

9) Match List I with List II :

	List-I		**List-II**
A.	Joseph Priestley	I	Discoverer of bacterial photosynthesis
B.	Jan Ingenhousz	II	Action spectrum of photosynthesis
C	T.W. Engelmann	III	Only the green part of the plant could release O_2
D	C.V. Neil	IV	Discoverer of O_2

Choose the correct answer from the options given below.

(1) A–I, B–II, C–III, D–IV

(2) A–III, B–IV, C–II, D–I

(3) A–IV, B–III, C–II, D–I

(4) A–IV, B–III, C–I, D–II

10) **Statement-I :-** Action spectrum resembles roughly the absorption spectra of chlorophyll-a and chlorophyll-b :-

Statement-II :- Most of the photosynthesis take place in the blue and red region of the spectrum.

(1) Both Statement-I and II are correct

(2) Statement-I is correct and II is not correct

(3) Statement-II is correct and I is not correct

(4) Both Statements are not correct

11) During creation of proton gradient in chloroplast :-
(i) pH of thylakoid lumen __(A)__
(ii) pH of stroma __(B)__
(iii) Concentration of H^+ ions in stroma __(C)__
(iv) Concentration of H^+ ions in thylakoid lumen __(D)__

(1) A, B → Increase; C, D → Decrease

(2) A, C → Increase; B, D → Decrease

(3) A, D → Increase; B, C → Decrease

(4) A, C → Decrease; B, D → Increase

12) Which of the following stages are best describing the Calvin cycle :-
(a) Carboxylation
(b) Oxidation
(c) Reduction
(d) Regeneration
(e) Decarboxylation
Choose the correct answer from the options given below :-

(1) (a), (b) and (c) only

(2) (a), (c) and (d) only

(3) (a), (b) and (d) only

(4) (d) and (e) only

13) **Assertion :** In C_4 plant photorespiration does not occur.
Reason : They have the mechanism that maintains the concentration of CO_2 at the site of RuBisCo enzyme.

(1) Both assertion and reason are correct and reason is correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not correct explanation of assertion.

(3) Assertion correct but reason is incorrect.

(4) Both assertion and reason are incorrect.

14) Which one of the following statement is not correct about photorespiration ?

(1) Photorespiration is the process that creates are important difference between C_3 and C_4 plants.

(2) The RUBP bind with O_2 to form one molecule of phosphoglycerate and phosphoglycolate in a pathway.

(3) In photorespiratory pathway there is no synthesis of ATP or NADPH.

(4) In photorespiration CO_2 is released without utilisation of ATP.

15) Which of the following represent the internal factors of plants ?
(a) Number of leaves (b) Size of leaves
(c) Age of leaves (d) Orientation of leaves
(e) Temperature (f) Mesophyll cells
(g) Sunlight (h) Amount of chlorophyll
(i) CO_2 concentration (Atmospheric)

(1) a, b, d only
(2) a, b, c, d, f, h only
(3) b, c, d, e, f only
(4) a, b, c, d, e, f only

16) (A) Photosynthesis is essentially a light dependent process, in which hydrogen from suitable oxidisable compound reduces the CO_2 into carbohydrates.
(B) Green substances in plants is located in special bodies in plant cells.
Name the researchers who stated the above statements.

(1) J.V. Sachs & Joseph priestly respectively
(2) T.W. Engelman & Van Neil respectively
(3) Van Neil & J.V. Sachs respectively
(4) Ingenbousz & J.V. Sachs respectively

17) From the statements given below choose the correct option :-
(A) The LHC are made up of pigments bound to proteins.
(B) C_4 plants are known to be more tolerant to high temperature than C_3 plants.
(C) Ultimate e^- acceptor in non-cyclic photophosphorylation is $NADP^+$.
(D) More the ATP used in photorespiration more is the photosynthesis.
(E) RuBisCO has a lower affinity for CO_2 when the CO_2 : O_2 in nearly equal.

(1) A, B, D are true
(2) A, B, C are true
(3) B, C, D are true
(4) A, D, E are true

18) In glycolysis the conversion of 2-phosphoenol pyruvic acid to pyruvic acid is?

(1) Energy utilising process
(2) Substrate level phosphorylation
(3) Isomerisation process
(4) Reduction process

19) Conversion of pyruvic acid into acetyl CoA takes place through enzyme :-

(1) Pyruvate decarboxylase
(2) Pyruvate kinase
(3) Pyruvate dehydrogenase
(4) Aldolase

20) In a coupled reaction GTP is converted to GDP with the simultaneous synthesis of ATP from ADP. This reaction is related with :

(1) Glycolysis
(2) Tricarboxylic acid cycle.
(3) Fermentation
(4) ETS

21) Which of the following complex contains cytochrome a, a_3 and two copper centres?

(1) Complex I
(2) Complex II
(3) Complex III
(4) Complex IV

22) _____ is the process which is associated with release and utilisation of the energy stored in $NADH{\cdot}H^+$ & $FADH_2$?

(1) Glycolysis
(2) Krebs cycle
(3) Link reaction
(4) ETS and oxidative phosphorylation

23) The Respiratory Quotient depends upon the :-

(1) Type of enzymes
(2) Type of organisms
(3) Type of minerals
(4) Type of respiratory substrate

24) During fermentation the energy obtained is _____ of energy of glucose.

(1) More than 7%
(2) less than 7%
(3) equal to
(4) Twice

25) Which of the following molecule is not considered an intermediate in the amphibolic pathway ?

(1) DHAP
(2) Pyruvate
(3) Acetyl CoA
(4) Carbondioxide

26) Which of the following statements are correct about glycolysis ?
A. Glucose undergoes partial oxidation to form two pyruvic acid

B. Sucrose is converted into glucose and fructose by invertase.
C. Glycolysis only occurs is nucleus of the cell.
D. Pyruvic acid is the key product of glycolysis
E. PGAL is oxidised with inorganic phosphate to get converted into 1-6 bisphosphate.
Choose the correct answer-

(1) A, B, D only

(2) B, C, D, E only

(3) A, C, D only

(4) A, B, D, E only

27) Which statement is **correct** ?

(1) During link reaction, two molecules of NADPH are produced from the metabolism of two molecules of pyruvic acid.

(2) Pyruvate undergoes oxidative decarboxylation when it enters in electron transport system

(3) Link Reaction is catalysed by pyruvate dehydrogenase through complex set of reactions.

(4) Pyruvate dehydrogenase require the participation of several coenzymes including $NADP^+$ and FMN.

28) Match the column I and column II

Column I (Chemical)		**Column II (Number of carbons)**	
A.	Acetyl group	(i)	2 carbons
B.	Citric acid	(ii)	5 carbons
C.	α ketoglutaric acid	(iii)	6 carbons
D.	Oxalo acetic acid	(iv)	4 carbons

(1) A-i,B-ii,C-iii,D-iv

(2) A-ii,B-iii,C-i,D-iv

(3) A-i,B-iii,C-ii,D-iv

(4) A-i,B-iii,C-iv,D-ii

29) Match the list I with list II :

	List I		**List II**
(A)	Complex I	(i)	Succinate dehydrogenase
(B)	Complex II	(ii)	NADH dehydrogenase
(C)	Complex IV	(iii)	ATP synthase
(D)	Complex V	(iv)	Cytochrome c oxidase

(1) A-ii, B-i, C-iv, D-iii

(2) A-i, B-iv, C-ii, D-iii

(3) A-i, B-ii, C-iii, D-iv

(4) A-ii, B-i, C-iii, D-iv

30) **Assertion :-** The presence of oxygen is vital in aerobic respiration.
Reason :- Oxygen derives the whole process by removing hydrogen from the system.

(1) Assertion is true & Reason is false

(2) Assertion is false & Reason is true

(3) Assertion & Reason both are true & Reason is not correct explanation of Assertion

(4) Assertion & Reason both are true & Reason is correct explanation of Assertion

31) Read the following terminology of column-I & II and identify their correct match :

Column-I (Substrates)		**Column-II (RQ values)**	
(A)	Carbohydrate	(i)	0.7
(B)	Protein	(ii)	1
(C)	Organic acid	(iii)	0.9
(D)	Fat	(iv)	–1
		(v)	0
		(vi)	4
		(vii)	Infinite

(1) A-v, B-iii, C-i, D-iv

(2) A-ii, B-i, C-vi, D-iii

(3) A-iii, B-i, C-iv, D-v

(4) A-ii, B-iii, C-vi, D-i

32) **Statement-I :** In both lactic acid and alcohol fermentation 13% of energy of glucose is released.
Statement-II : Lactic acid and alcohol fermentation are hazardous for cells.
Choose the correct option :

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct

33) **Statement-I** : Fats would need to be broken down into glycerol and fatty acids first. If fatty acids were to be respired they would be degraded to PGAL and enter the pathway. Glycerol would enter the pathway after being converted to acetyl Co-A.
Statement-II : Glucose is the favoured substrate for respiration thats why all carbohydrates are usually first converted into glucose before they are used for respiration.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

34)

Match List-I with List-II –

List-I		**List-II**	
(a)	Developmental Heterophylly	(I)	Growth and Differentiation
(b)	Environmental Heterophylly	(II)	Cotton
(c)	Development	(III)	Buttercup

Choose the correct answer form the options given below :

(1) (a) – (II), (b) – (III), (c) – (I)

(2) (a) – (III), (b) – (II), (c) – (I)

(3) (a) – (I), (b) – (II), (c) – (III)

(4) (a) – (III), (b) – (I), (c) – (II)

35) Match the column-I with column-II.

	Column-I		**Column-II**
(a)	Maize root apical meristem growth	(i)	Increase in surface area
(b)	Watermelon cell growth	(ii)	Increase in length
(c)	Growth of pollen tube	(iii)	Increase in size of cell
(d)	Growth in dorsiventral leaf	(iv)	Increase in cell number

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(ii)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

36) The volatile substance is released from ripened oranges that __A__ the ripening of stored un-ripened banana. It was identified as __B__.

(1) A-Reduces, B-Abscisic acid

(2) A-Hastened, B-Ethylene

(3) A-Stop, B-Auxin

(4) A-Hastened, B-Cytokinins

37) Select the incorrect statements :
(a) Intrinsic factor that control the development in plant include both intracellular or intercellular factors.
(b) Intercellular factors are also known as genetic factors.
(c) Chemicals such as plant growth regulators are intracellular factors always.
(d) Temperature, water, oxygen and nutrition are extrinsic factors that control the development in plants.

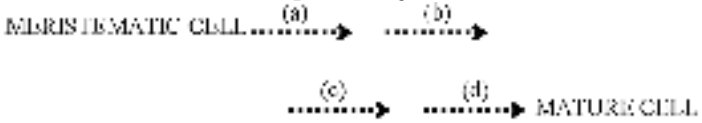
(1) a and d

(2) b and c

(3) a, b and c

(4) c and d

38) Fill the blanks respectively :



(1) (a) – Plasmatic growth, (b) – Differentiation,
(c) – Elongation, (d) – Maturation

(2) (a) – Differentiation, (b) – Elongation,
(c) – Plasmatic growth, (d) – Maturation

(3) (a) – Differentiation, (b) – Plasmatic growth,
(c) – Maturation, (d) – Elongation

(4) (a) – Differentiation, (b) – Plasmatic growth,
(c) – Elongation, (d) – Maturation

39) E. Kurosawa reported symptoms of foolish seedling disease in :–

(1) Canary grass

(2) Rice

(3) Oat

(4) Orange

40) Match the column A with column B and select the correct option from given below :

	Column A		**Column B**
a	Auxins	i	Antagonist of GA_s
b	Gibberellines	ii	Overcome apical dominance
c	Cytokinins	iii	Speed up malting process in brewing industry
d	ABA	iv	Promote flowering in pineapple

(1) a-iv, b-iii, c-ii, d-i

(2) a-ii, b-iii, c-i, d-iv

(3) a-iv, b-ii, c-iii, d-i

(4) a-ii, b-iv, c-iii, d-i

41) Which option from given below, explains that GA_3 is beneficial for those farmers who want to extend market period of crop.

(1) Increase the length of grape stalk

(2) Speed up the malting process

(3) Increase the length of stem of sugarcane

(4) Delay senescence in fruits

42) ABA is called as the stress hormone because :

(1) Stimulate closure of stomata

(2) inhibit leaf fall

(3) Play role in germination

(4) Play role in induction of flowering

43) Auxin was first isolated from ?

(1) Human urine

(2) Pineapple

(3) Coconut

(4) Coconut leaves

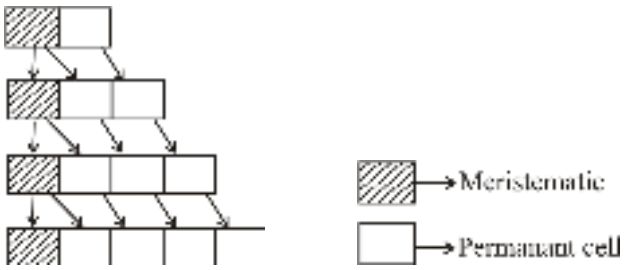
44) Which equation represents arithmetic growth?

(1) $L_t = L_0e^{rt}$

(2) $L_0 = L_te^{rt}$

(3) $L_t = L_0 + rt$

(4) $W_t = W_0e^{rt}$

45) 

Above figure represents :-

(1) Zygote division

(2) Arithmetic growth

(3) Geometric growth

(4) Both (1) & (2)

46) During depolarisation of nerve :-

(1) Na^+ VGC – Close, K^+ VGC – open

(2) Na^+ VGC – Open, Na^+-K^+ pump – open

(3) K^+ VGC – Close, Na^+-K^+ pump – close

(4) Na^+ VGC – Open, K^+ leak channel close

47) Schwann cells are found on :-

(1) Myelinated nerve fibres

(2) Unmyelinated nerve fibres

(3) Cell body of neuron

(4) Both (1) and (2)

48) **Assertion :** Electrical current can flow directly from one neuron into the other across electrical synapse.
Reason : Electrical synapse pre and post neuron are in very close proximity.

(1) Assertion and Reason both are correct and Reason is the correct explaination of Assertion

(2) Assertion and Reason both correct but Reason is not the correct explaination of Assertion

(3) Assertion is correct but Reason is incorrect

(4) Both Assertion & Reason are incorrect

49) Which of the following statement is incorrect about neuron ?

(1) Cyton is present in neuron

(2) Nissel's granules are present in Dendron, cell body and axon

(3) Axon is the longest cell process of neuron

(4) Dendrites transmit impulses towards the cell body

50) Match the following structures in the human brain with its function :

	Column-I		**Column-II**
(a)	Association area	(i)	eating and drinking urge
(b)	Thalamus	(ii)	Coordination between sensory and motor signaling
(c)	Medulla	(iii)	Cardiovassular reflexes
(d)	Hypothalamus	(iv)	Memory

(1) a-(ii), b-(iii), c-(iv), d-(i)

(2) a-(ii), b-(iii), c-(i), d-(iv)

(3) a-(iii), b-(ii), c-(iv), d-(i)

(4) a-(iv), b-(ii), c-(iii), d-(i)

51) **Statement-I :** The neural system and the endocrine system jointly co-ordinate and integrate all the activities of the organs so that they function in a synchronised fashion
Statement-II : The neural system provides an organised network of point to point connections for a very slow coordination.

(1) Both statement-I and statement-II are incorrect
(2) Statement-I is correct but statement-II is incorrect
(3) Statement-I is incorrect but statement-II is correct
(4) Both statement-I and statement-II are correct

52) Which one of the following is not the function of limbic system ?

(1) Regulation of sexual behaviour
(2) Expression of emotional reaction
(3) Languages comprehension
(4) Motivation

53) Which of the following statements or structures is not correct about the midbrain ?

(1) Has a canal (Cerebral aqueduct)
(2) Located between the thalamus and pons
(3) Involved in motivation
(4) Its dorsal part consists of four lobes (corpora quadrigemina)

54) Match the column & choose correct set of option :

	Column I	**Column II**	**Column III**
a.	Unipolar neuron	embryonic neuron	 Unipolar

b.	Multipolar neuron	Cerebral cortex	 Multipolar
c.	Bipolar neuron	Retina of eye	 Bipolar

(1) a, b

(2) b, c

(3) a, c

(4) a, b, c

55) During rise in stimulus which ion permeability is induced in neuron and it is extremly short lived?

(1) K^+

(2) Ca^{+2} & K^+

(3) Na^+ & K^+

(4) Na^+

56) Repolarisation of neuron is occured due to :-

(1) Influx of Na^+

(2) Influx of K^+

(3) Efflux of Na^+

(4) Efflux of K^+

57)

Glucocorticoids stimulates :-
(A) Gluconeogenesis (B) Lipolysis

(C) Proteolysis

(1) Only A and B

(2) Only A
(3) Only B
(4) All A, B and C

58) Which statement is incorrect regarding pineal gland ?

(1) Located on the ventral side of fore brain.
(2) Secretes a hormone called melatonin.
(3) It influences metabolism.
(4) Influence our defense capability.

59) Which of the following gastro-intestinal tract hormone, act on exocrine pancrease and stimulate secretion of water and bicarbonate ions :-

(1) Gastrin
(2) Secretin
(3) Cholecystokinin
(4) Gastric inhibitory peptide

60) Which one is not a group of antagonistic Hormones–

(1) MSH & ACTH
(2) Melatonin and MSH
(3) Insulin and glucagon
(4) Calcitonin and PTH

61) All of the following are gastrointestinal hormones except -

(1) Gastrin
(2) Secretin
(3) Cholecystokinin
(4) Erythropoitin

62) ACTH stimulates the synthesis and secretion of.....hormones called.....from adrenal cortex :-

(1) Protein, sex corticoids
(2) Steroid, mineralocorticoids
(3) Short peptides, mineralocorticods
(4) Steroid, glucocorticoids

63) **Assertion:** The hormones produced by hypothalamus are of two types, the releasing hormones and the inhibiting hormones.
Reason: A hypothalmic hormone called Gonadotrophin releasing hormone(GnRH) stimulates the pituitary gland to synthesis and release of gonadotrophins. On the other hand, somatostatin from the hypothalamus inhibit the growth hormone from the pituitary.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

64) Read the following statements (A-D) (A) Noradrenaline and adrenaline are called as catecholamines. (B) Parathyroid hormone stimulates reabsorption of Ca^{+2} by the renal tubules and increases Ca^{+2} absorption from the digested food (C) Maintenance of water and electrolyte balance is also influenced by thyroid hormones. (D) Thyroid gland also secretes a thyrocalcitonin which regulates the blood calcium levels. How many of the above statements are true :-

(1) Three

(2) Four

(3) Two

(4) One

65) Choose the correct answer among the following options :-

Hormones		**Function**	
(A)	Epinephrine	(i)	Increase in muscle growth
(B)	Testosterone	(ii)	Decrease in blood pressure
(C)	Glucagon	(iii)	Decrease in liver glycogen content
(D)	Atrial natriuretic factor	(iv)	Increase heart beat

(1) A-iv, B-i, C-ii, D-iii

(2) A-iv, B-i, C-iii, D-ii

(3) A-i, B-ii, C-iii, D-iv

(4) A-i, B-iv, C-ii, D-iii

66) **Assertion :** Underproduction of hormones by adrenal cortex alters carbohydrate metabolism.
Reason : Glucocorticoids which are released from zona fasciculata (A part of adrenal cortex) regulate carbohydrate metabolism.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

67) Out of the following given hormones how many function with the help of secondary messenger ?
Insulin, Epinephrine, F.S.H, Glucagon, Testosterone, Estradiol

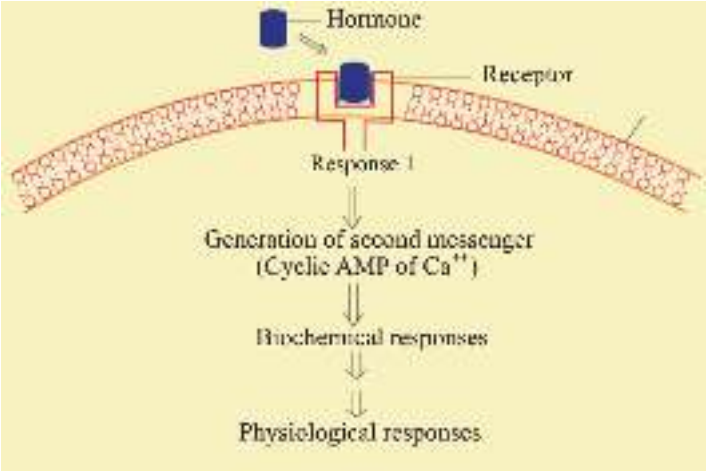
(1) Three

(2) Two

(3) Four

(4) One

68) Below diagram represent which of the following hormone?



(1) Cortisol
(2) Aldosterone
(3) Oestrogen
(4) Parathormone

69) **Statement-I :-** JG cells of the kidney produce erythropoietin hormone.
Statement-II :- ANF is introduced in circulation when blood pressure increases.

(1) Statement I & II correct
(2) Statement I & II incorrect
(3) Statement I correct & II incorrect
(4) Statement I incorrect & II correct

70) The effect caused by Non-functioning of Neurohypophysis ?

(1) BMR increased
(2) Hyperglycaemia
(3) Diabetes insipidus
(4) Hypoglycaemia

71) Hypersecretion of which hormone causes Grave's disease ?

(1) Aldosteron
(2) Adrenalin
(3) Thyroxine
(4) Calcitonin

72) Corpus luteum secretes a hormone called :-

(1) Prolactin
(2) Progesterone
(3) Aldosterone
(4) Testosterone

73) Which part of body secretes anti-inflammatory and emergency hormone respectively?

(1) Thymus and adrenal cortex
(2) Adrenal cortex and adrenal medulla
(3) Pineal gland and adrenal medulla
(4) Adrenal medulla and adrenal cortex

74) Which is not the function of aldosterone ?

(1) Maintainance of body fluid volume
(2) Myelination of nerve fibre
(3) Reabsorption of Na^+ and H_2O
(4) Excretion of K^+ and PO_4^{-3}

75) Which of the following hormone does not play a significant role in erythropoiesis?

(1) Testosterone
(2) Cortisol
(3) Iodothyronines
(4) Parathyroid hormone

76) Which of the following is set of steroidal hormones ?

(1) Estrogen, Progesterone, Prolactin
(2) Estrogen, Progesterone, Cortisol
(3) Estrogen, ADH, Cortisol
(4) Estrogen, Progesterone, Oxytocin

77) What is the type of joint between ribs and sternum?

(1) Cartilagenous joint
(2) Synovial joint
(3) Fibrous joint
(4) Angular joint

78)

ATPase enzyme, needed for muscle contraction is located in :

(1) Troponin
(2) Tropomyosin
(3) Actin
(4) Myosin

79) Which statements are incorrect ?
(A) During muscle contraction A-band decreases in length.

(B) Red muscle fibers contains high quantity of myoglobin.
(C) Z-line bisects the I-band.
(D) Length of sarcomere increase during muscle contraction.

(1) A and D
(2) B and D
(3) A and C
(4) C and D

80) Which is not true for scapula in the following?

(1) It is found between 2nd to 7th ribs
(2) It has a cavity which is found above the acromian process
(3) It is found in dorsal part of throrax
(4) It is a flat triangular bone

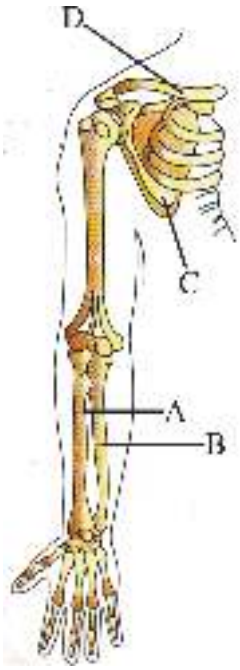
81) Choose the correct matching :-

	Synovial Joint		**Bones Involved**
(a)	Ball and socket	(i)	Carpal and metacarpal of thumb
(b)	Hinge	(ii)	Atlas and axis
(c)	Pivot	(iii)	Frontal and parietal
(d)	Saddle	(iv)	Knee
		(v)	Humerus and pectoral girdle

(1) a-v, b-iv, c-ii, d-i
(2) a-ii, b-v, c-iv, d-i
(3) a-i, b-ii, c-v, d-iv
(4) a-v, b-iv, c-iii, d-i

82) Which one is odd from the following ?

(1) Elbow joint
(2) Knee joint
(3) Wrist joint
(4) Pubic symphysis



83) In given diagram A,B,C and D respectively represents :-

(1) Radius, Ulna, Scapula, Clavicle
(2) Ulna, Radius, Scapula, Clavicle
(3) Radius, Ulna, Clavicle, Scapula
(4) Ulna, Radius, Clavicle, Scapula

84) Intercalated discs are characterstics of muscles that are found in :-

(1) Brain
(2) Spinal cord
(3) Stomach
(4) Heart

85) Structurally Heart muscle is similar to -

(1) Skeletal muscle
(2) Smooth muscle
(3) Visceral muscle
(4) Involuntary muscle

86) Following are the events of muscular contraction, arrange them in correct order?
(A) Action potential generates in the sarcolemma
(B) Hydrolysis of ATP
(C) Myosin head pulls the attached actin filaments
(D) Release of calcium ions into the sarcoplasm
(E) Shortening of sarcomere

(1) A → B → C → D → E
(2) A → C → D → B → E
(3) A → D → B → C → E
(4) A → C → B → D → E

87) **Statement-I :** Red muscle contain plenty of mitochondria which can utilise the large amount of O_2 for ATP production.
Statement-II : Red muscles depend on anaerobic process for energy.

(1) Statement-I is correct but Statement-II is incorrect.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Both Statement are correct.

(4) Both Statement are incorrect.

88) **Assertion (A) :** Synovial joint help in locomotion and many other movement.
Reason (R) : Synovial joints are characterised by the presence of a fluid filled synovial cavity between the articulating surface of the two bones. Such an arrangement allow considerable movement.

(1) Both A and R are correct and R is correct explanation of A.

(2) Both A and R are correct and R is not correct explanation A

(3) A is true and R is wrong

(4) Both A and R are wrong

89) Identify the statements as true (T) or false (F) and choose the correct option ?
(i) Cilia help in locomotion in *Paramecium*
(ii) *Hydra* uses tentacles for capturing of prey and also for locomotion
(iii) Female reproductive tract in humans exhibit flagellar movement to propel the ovum
(iv) Macrophages and leucocytes of humans are specialised cells that exhibit amoeboid movement by forming pseudopodia

	i	ii	iii	iv
(1)	T	T	F	F
(2)	T	F	T	F
(3)	T	T	F	T
(4)	T	T	T	F

(1) 1

(2) 2

(3) 3

(4) 4

90)

Head of the humerus articulates with :-

(1) Acetabulum of pectoral girdle.

(2) Acetabulum of pelvic girdle.

(3) Glenoid cavity of pectoral girdle.

(4) Glenoid cavity of pelvic girdle.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	2	3	1	2	3	3	3	3	1	2	2	2	4	3	1	4	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	3	3	2	2	4	2	1	1	2	4	2	2	1	3	1	4	2	4
Q.	41	42	43	44	45															
A.	2	1	1	3	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	4	1	1	4	4	3	3	1	2	3	3	3	1	2	1	2	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	3	4	2	1	2	3	3	1	4	3	1	3	4	2	1	3	3	3
Q.	86	87	88	89	90															
A.	1	3	2	1	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	1	4	1	1	4	2	3	1	4	2	1	4	2	3	2	2	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	4	2	4	1	3	3	1	4	4	4	3	1	1	2	2	1	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	1	3	2	3	4	1	2	4	2	3	3	4	4	4	4	1	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	1	2	2	1	3	4	1	3	3	2	2	2	4	2	1	4	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	1	4	1	3	1	1	3	3										

SOLUTIONS

PHYSICS

1)

SHM relations: $v^2 = \omega^2(A^2 - x^2)$, $a = -\omega^2 x$

Given A = 2 cm, x = 1 cm. Condition $|\vec{a}| = |\vec{v}|$.

$|\vec{a}| = \omega^2 |\vec{x}|, \quad |\vec{v}| = \omega\sqrt{A^2 - x^2}$.

Equating for ω

$\Rightarrow \omega^2 x = \omega\sqrt{A^2 - x^2}$

$\omega x = \sqrt{A^2 - x^2}$.

Square both sides:

$\omega^2 x^2 = A^2 - x^2$.

Thus

$$\omega^2 = \frac{A^2 - x^2}{x^2} = \frac{(2)^2 - (1)^2}{(1)^2} = \frac{4-1}{1} = 3$$

So $\omega = \sqrt{3}$rad/s. Frequency $f = \frac{\omega}{2\pi} = \frac{\sqrt{3}}{2\pi}s^{-1}$

2)

Spring stretched by mass M :

$$k = \frac{Mg}{\ell}$$

After adding tiny mass m, total oscillating mass : M + m.

Time period :

$$T = 2\pi\sqrt{\frac{M+m}{k}} = 2\pi\sqrt{\frac{(M+m)\ell}{Mg}}$$

3)

(A) in parallel

$K_{eq} = K_1 + K_2$

$T = 2\pi\sqrt{\frac{m}{K_1 + K_2}} \Rightarrow (s)$

(B) in parallel $K_{eq} = K + K = 2K$

$T = 2\pi\sqrt{\frac{m}{2k}} \Rightarrow (r)$

(C) in series $K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$

$T = 2\pi\sqrt{\frac{m(K_1 + K_2)}{K_1 K_2}} \Rightarrow (p)$

(D) in series $K_{eq} = \frac{K\,K}{K+K} = \frac{K}{2}$

$T = 2\pi\sqrt{\frac{2m}{K}} \Rightarrow (q)$

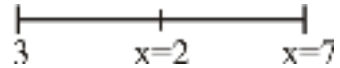
4)

$U = 2 - 20x + 5x^2$

$F = -\frac{dU}{dx} = -(-20 + 10x)$

$F = 20 - 10x$

x = 2 M.P.



A = 5 cm x_{max} = 7 cm

5)

Acceleration proportional to displacement (a ∝ x) is not sufficient; it must be a ∝ –x . The reason mentions the direction of displacement is considered in SHM, which explains why displacement proportionality is insufficient.

6)

$T = 2\pi\sqrt{\frac{\ell}{g}}$; $T' = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$

$g_{eff} = g - \frac{g}{\sigma} = g\left(1 - \frac{1}{\sigma}\right)$

$\frac{T'}{T} = \sqrt{\frac{\sigma}{\sigma - 1}}$

7)

$U=4(1 - \cos 2x)$

$\because \quad F = -\frac{dU}{dx} \Rightarrow F = -8 \sin 2x$

For small oscillations, x will be small hence

$F = -8(2x) = -16x \quad \Rightarrow k = 16$ and m = 1kg

$\therefore \quad \omega^2 = \frac{k}{m} = \frac{16}{1} = 16 \Rightarrow \omega = 4$

$\Rightarrow \quad T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = \frac{\pi}{2}$

8)

$E_1 = \frac{1}{2}Kx^2 \Rightarrow x = \sqrt{\frac{2E_1}{K}}$ ...(i)

and

$E_2 = \frac{1}{2}Ky^2 \Rightarrow y = \sqrt{\frac{2E_2}{K}}$ ...(ii)

Now

$E_3 = \frac{1}{2}K(x + y)^2$

$= \frac{1}{2}K\left(\sqrt{\frac{2E_1}{K}} + \sqrt{\frac{2E_2}{K}}\right)^2$

$= \frac{1}{2}K\left(\sqrt{\frac{2}{K}}\left(\sqrt{E_1} + \sqrt{E_2}\right)\right)^2$

$= \frac{1}{2}K \times \frac{2}{K}\left(\sqrt{E_1} + \sqrt{E_2}\right)^2 = \left(\sqrt{E_1} + \sqrt{E_2}\right)^2$

$= E_1 + E_2 + 2\sqrt{E_1E_2}$

9)

$K.E = \frac{1}{n}P.E$

$\frac{1}{2}K(A^2 - x^2) = \frac{1}{n}\frac{1}{2}kx^2$

$nA^2 - nx^2 = x^2$

$x^2(1 + n) = nA^2$

$x = \sqrt{\frac{n}{n+1}}.A$

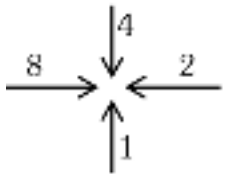
10) $\omega^2 A = \omega A$

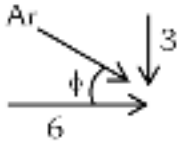
$\omega = 1$

$\frac{2\pi}{T} = 1$

$T = 2\pi = 6.28$ second.

11)





$A_r = \sqrt{3^2 + 6^2} = \sqrt{45}$; $\tan\phi = \frac{3}{6} \Rightarrow \phi = \tan^{-1}\left(\frac{1}{2}\right)$

12)

	A	B
Frequency	n_A	n_B
Tension	T	T
Length	2L	3L
Radii	3r	2r

Vibration Fundamental $n_A = \frac{1}{2(2L)}\sqrt{\frac{T}{\rho\pi(3r)^2}}$

Second Overtone $n_B = \frac{3}{2(3L)}\sqrt{\frac{T}{\rho\pi(2r)^2}}$

$\frac{n_A}{n_B} = \frac{1}{3}$

13) Equation of the component wave are
$y = A \sin(\omega t - kx)$ and $y = A \sin(\omega t + kx)$ where ; $\omega t - kx$ = constant or $\omega t + kx$ = constant
Differentiating w.r.t 't';

$\omega - k\frac{dx}{dt} = 0$ and $\omega + k\frac{dx}{dt} = 0$

$\Rightarrow v = \frac{dx}{dt} = \frac{\omega}{k}$ and $v = -\frac{\omega}{k}$

i.e. ; the speed of component waves is $\left(\frac{\omega}{k}\right)$

14) As x = 0 is node (Reflection from rigid end) ⇒ standing wave should be $y = 2a \sin kx \sin \omega t$
$y_i = a \cos(kx - \omega t)$
$y_r = a \cos(kx + \omega t + \pi)$
$y_r = -a \cos(kx + \omega t)$

15) First, find the ratio of the given frequencies: 255:425:595 To simplify this ratio, we find the greatest common divisor (GCD) of these numbers, which is 85. Dividing each frequency by 85 gives the ratio: 3:5:7 Since the pipe is only producing **odd harmonics** (3rd, 5th, 7th), it must be a **closed pipe**.

$3.\frac{V}{4\ell} = 255$

$\frac{V}{4\ell} = 85$ Hz

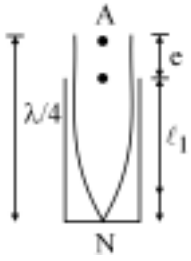
Fundamental frequency

16) $V_s \propto \sqrt{T} \Rightarrow \frac{V_{s_1}}{V_{s_2}} = \sqrt{\frac{T_1}{T_2}}$

$\Rightarrow \frac{v}{\sqrt{5}v} = \sqrt{\frac{T}{T+800}}$

$\Rightarrow \frac{1}{5} = \frac{T}{T+800} \Rightarrow T = 200$ K = – 73°C

17)



$\ell_1 + e = \frac{\lambda}{4}$

$\ell_1 = \frac{\lambda}{4} - e$

18) Maximum particle velocity = Aω

wave velocity = $\frac{\omega}{k}$

$\frac{Vp_{max}}{V_w} = AK.$

in equation

$y = 3 \sin \frac{\pi}{2}(50t - x)$

$y = 3 \sin \left(25\pi t - \frac{\pi}{2}x\right)$

Compare it with

$y = A \sin (\omega t - kx)$

$AK = \frac{3\pi}{2}$

19)

$A\omega = \frac{V}{10}$

$A\omega = \frac{10}{10}$

$A\omega = 1$

$10^{-3}\omega = 1$

$\omega = 10^3$

$\Rightarrow f = \frac{10^3}{2\pi}Hz.$

$V = f\lambda$

$10 = \frac{10^3}{2\pi}\lambda$

$\lambda = 2\pi \times 10^{-2}$ m

20) wave velocity = $\frac{\text{coefficient of t}}{\text{coefficient of x}} = \frac{60}{2} = 30\ ms^{-1}$

Also coefficient of t and x are of same sign

⇒ so it propagation in –x direction

21)

Transverse + mechanical wave
⇒ (solid string) & (water ripple)

22) In hydrogen gas,

$$V_{H_2} = \sqrt{\frac{\gamma_{H_2} RT}{m_{H_2}}} = \sqrt{\frac{\frac{7}{5}RT}{2}} = \sqrt{\frac{7\,RT}{10}} = v_0 \quad \ldots(1)$$

Let mixture contains x moles each of H_2 & He

$$\gamma_{mix} = \frac{n_1 C_{p_1} + n_2 C_{P_2}}{n_1 C_{v_1} + n_2 C_{v_2}}$$

$$\gamma_{mix} = \frac{x \times \frac{7R}{2} + x \times \frac{5R}{2}}{x \times \frac{5R}{2} + x \times \frac{3R}{2}} = \frac{3}{2}$$

$$m_{mix} = \frac{x \times 2 + x \times 4}{x + x} = 3$$

 velocity of sound in mixture,

$$v_{mix} = \sqrt{\frac{\gamma_{mix} RT}{m_{mix}}} = \sqrt{\frac{\frac{3}{2}RT}{3}}$$

$$v_{mix} = \sqrt{\frac{RT}{2}} \quad \ldots(2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{v_0}{v_{mix}} = \frac{\sqrt{\frac{7}{10}RT}}{\sqrt{\frac{RT}{2}}} = \sqrt{\frac{14}{10}} = \sqrt{\frac{7}{5}}$$

$$v_{mix} = \sqrt{\frac{5}{7}} v_0$$

23) We are given:

A. $L_1 = 0.750$ m,
B. $L_2 = 0.770$ m,
C. Beat frequency $\Delta n = 3$ beats/sec.

Using the formula for beat frequency:

$$\Delta n = \frac{v}{4}\left|\frac{1}{L_1} - \frac{1}{L_2}\right|$$

Substitute the values for L_1 and L_2:

$$3 = \frac{v}{4}\left|\frac{1}{0.750} - \frac{1}{0.770}\right|$$

Calculate the difference:

$$\frac{1}{0.750} - \frac{1}{0.770} \approx 1.3333 - 1.2987 = 0.0346$$

Now solve for v:

$$3 = \frac{v}{4} \times 0.0346$$

$$v = \frac{3 \times 4}{0.0346} \approx \frac{12}{0.0346} \approx 346.5\text{m/s}$$

24) A' = 0.9 A_0 in 5s
in next 10s so total time become 15s, 3
time of 5 sec so

$A'' = (0.9)^3 A_0$

$= 0.729\ A_0$

Alternate Solution

$A = A_0 e^{-\lambda t}$

$A_1 = 0.9\ A_0 = A_0\ e^{-5\lambda}$ .....(i)

$\Rightarrow e^{-5\lambda} = 0.9$

After next 10s total time = 5 +10 = 15s

$A_f = A_0 e^{-15\lambda} = A_0 (e^{-5\lambda})^3$

$A_f = A_0 (0.9)^3 = 0.729\ A_0$

If $A_f = \alpha A_0$ then $\alpha = 0.729$

25) $e = \dfrac{L_2 - 3L_1}{2} = \dfrac{32-30}{2} = 1\ cm$

26) $Q_{Block} = Q_{Ice}$

$m_B s_B \Delta T = mL$

$1200 \times 0.1 \times 1000 = m \times 80$

$m = \dfrac{1200 \times 100}{80}$

$m = 1500\ gm$

$m = 1.5\ kg$

27) Isothermal process, T = constant

$PV = nRT$ = constant

$P_1V_1 = P_2V_2$

$PV = P_A(V/8)$

$P_A = 8P$

Adiabatic process, PV^γ = constant

γ for monoatomic gas is $\dfrac{5}{3}$.

$P_1V_1^\gamma = P_2V_2^\gamma$

$\dfrac{P_B}{P} = \left(\dfrac{V_1}{V_2}\right)^\gamma = \left(\dfrac{V}{V/8}\right)^{\frac{5}{3}}$

$P_B = 32P$

$\dfrac{P_B}{P_A} = \dfrac{32P}{8P} = 4$

28) $dU = nC_v dT$

$dQ = nC_p dT$

$dW = PdV = nRdT$ (isobaric process)

$dU : dQ : dW \ : \ C_v : C_p : R$

$= \dfrac{5R}{2} : \dfrac{7R}{2} : R = 5 : 7 : 2$

29) $W = \int PdV = \int \dfrac{nRT}{(V-b)} dV$

$= nRT \ln(V-b)\Big|_{V}^{2V}$

$W = nRT \ln\left(\frac{2V-b}{V-b}\right)$

30) VT = K

$\Rightarrow V\left(\frac{PV}{nR}\right) = k \Rightarrow PV^2 = K$

$C = \frac{R}{1-x} + C_V$ (For polytropic process)

$C = \frac{R}{1-2} + \frac{3R}{2} = \frac{R}{2}$

$\therefore \Delta Q = nC\,\Delta T$

$= \frac{R}{2} \times \Delta T$

31) For any scale

$\frac{x - LFP}{UFP - LFP} = \text{const.}$

$\frac{\frac{x_0}{2} - \frac{x_0}{3}}{x_0 - x_0/3} = \frac{C-0}{100-0}$

$\frac{\frac{x_0}{6}}{\frac{2x_0}{3}} = \frac{C}{100}$

C = 25°C

32)

In process CA $\Rightarrow PV^{-2}$ = constant

$\Rightarrow TV^{-3}$ = constant

$T_A = T_C \times \left(\frac{V_A}{V_C}\right)^3 = 2400K$

$W_{CA} = \frac{nR(T_A - T_C)}{1-(-2)} = 700R$

In process AB (Isobaric),

$T_B = T_A \times \frac{V_B}{V_A} = 1200K$

$W_{AB} = nR(T_B - T_A) = -1200R$

$\Delta U_{BC} = nC_V(T_C - T_B) = -2250R$

$Q_{AB} = nC_P(T_B - T_A) = -4200R$

33) $H = \frac{KA(T_H - T_C)}{d} \Rightarrow A = \frac{Hd}{K(T_H - T_C)}$

$= \frac{100 \times 0.20}{45 \times 20} = 0.022\ m^2$

34)

AD → $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$P_2 = P(2)^{\gamma} = 2^{1.4}\ P = 2.64\ P$

35) In AD process $\left(\gamma = \frac{5}{3}\right)$

$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = \left(\frac{6}{2}\right)^{5/3} = (3)^{5/3} = 6.19$

$P_2 = 6.19 \times 10^5\ N/m^2$

36) $\frac{Q}{t} = \sigma e A T^4$

Ratio $\frac{(Q/t)_1}{(Q/t)_2} = \frac{A_1T_1^4}{A_2T_2^4} = \frac{r_1^2T_1^4}{r_2^2T_2^4}$

$= \frac{(1)^2}{(4)^2} \times \left(\frac{4 \times 10^3}{2 \times 10^3}\right)^4 = 1 : 1$

37)

$\because I = \frac{ML^2}{12}$

$\Rightarrow I \propto L^2$

$\Rightarrow \frac{\Delta I}{I} = 2\frac{\Delta L}{L}$

[$\because \Delta L = L\alpha\Delta T$]

$\Rightarrow \frac{\Delta I}{I} = 2\frac{L\alpha\Delta T}{L} = 2\alpha\,\Delta t$

38) $\frac{\Delta C}{5} = \frac{\Delta F}{9}$

$\Delta F = \frac{9}{5} \times 30 = 54^{o}$

39)

FLOT is based upon energy conservation whereas SLOT is based upon entropy.

40) $\eta = 1 - \frac{T_2}{T_1}$

Case I :-

$\frac{50}{100} = 1 - \frac{(7 + 273)}{T_1}$

$T_1 = 560\ K$

Case II :-

$$\frac{70}{100} = 1 - \frac{280}{T'_1}$$
$$T'_1 \simeq 933\,K$$
$$\Delta T = T'_1 - T_1 = 373\,K$$

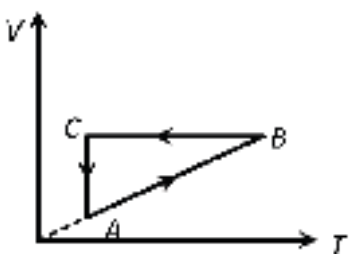
41) Heat gained by
Calorimeter = [x × (35 – 30)]
By water (Heat gain) ⇒ $0.2 \times 4.2 \times 10^3 \times (35–30)$
Heat Loss by water = $0.1 \times 4.2 \times 10^3 \times (60–35)$
Total Heat lost = Total heat gain
$0.1 \times 4.2 \times 10^3 \times (60–35)$
$= 0.2 \times 4.2 \times 10^3 \times (35–30) + x(35–30)$
(x = 1260 J/k)

42)

As $\frac{dQ}{dt} = ms\frac{dT}{dt} \Rightarrow \frac{dT}{dt} = \frac{1}{ms}\frac{dQ}{dt}$
where $\frac{dQ}{dt} = \sigma A(T^4 - T_s^4)$ = same for both
Now, $\frac{dT}{dt} \propto \frac{1}{m}$
$m_{hollow} < m_{solid}$ for same radius & material
$\left(\frac{dT}{dt}\right)_{Hollow} > \left(\frac{dT}{dt}\right)_{Solid}$
So, hollow sphere will cool at a faster rate for all values of T.

43)

rms speed $v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2}{3}}$
$= \sqrt{\frac{1^2 + 1^2 + 1^2}{3}} = 1$ km s^{-1}
Average speed $v_{av} = \frac{v_1 + v_2 + v_3}{3}$
$= \frac{1 + 1 + 1}{3} = 1$ km $s^{-1} \Rightarrow \frac{v_{rms}}{v_{av}} = \frac{1}{1} = 1$

44) 

From A to B
V ∝ T
So pressure is constant, volume increases so graph may be any of option (1) & (3)

For B to C, volume is constant and temperature decreases, $P \propto T$
So pressure decreases option (1) is not possible option (3) is possible
For C to A
Temperature is constant,
Volume decreases
PV = constant
So only option (3) is correct.

45) $17 + x = \frac{\lambda}{4}$ ...(1)

$53 + x = \frac{3\lambda}{4}$ ...(2)

(where x = end correction)

(2) – (1) ⇒ $\frac{\lambda}{2} = 36 \Rightarrow \lambda = 72$ cm

$\frac{(2)}{(1)} \Rightarrow x = 1$ cm

Third resonance length + x = $\frac{5\lambda}{4}$

Third resonance length = $\frac{5\lambda}{4} - x = \frac{5 \times 72}{4} - 1$

= 89 cm

CHEMISTRY

46)

Theoretical

47) $\frac{\text{Lassigne's test}}{\text{Blood Red colour}} \rightarrow \boxed{\text{N and S}}$

48)

Theoretical

49)

Theoretical

50) • Mass of organic compound = 0.15 g
• Mass of AgBr formed = 0.12 g
• Atomic mass of Br = 80 g/mol
• Molar mass of AgBr = 188 g/mol
Calculation :
1. Mass of Bromine in AgBr:

Mass of Br = (Atomic mass of Br / Molar mass of AgBr) × Mass of AgBr
Mass of Br = (80 / 188) × 0.12 g ≈ 0.051 g

2. Percentage of Bromine (% Br):
% Br = (Mass of Br / Mass of organic compound) × 100
% Br = (0.051 g / 0.15 g) × 100 ≈ 34%

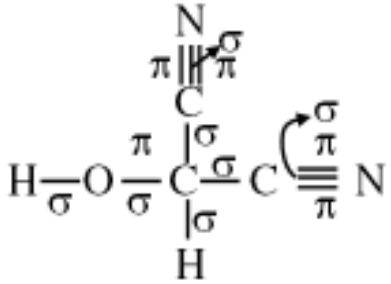
Final Answer :
Option (1)

51)

Theoretical

52)

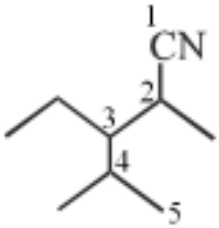
Explanation the Question:
The number σ and π bond is the given compound respectively.
Concept:
This question is based on σ and π calculation.
Solution:



7σ, 4π
Final Answer: (4)

53)

Fact

54) 

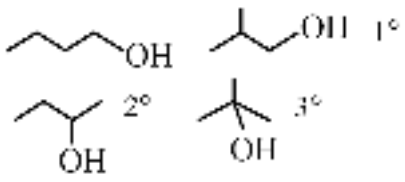
55)

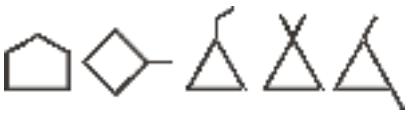
Theoretical

56) Fact

57)

Theoretical

58) 

59)  5 isomer

60)

Theoretical

61)

Theoretical

62)

Theoretical

63)

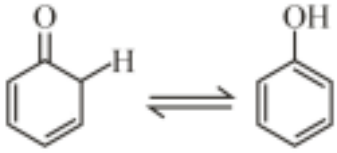
Position of substituent changes.

64)

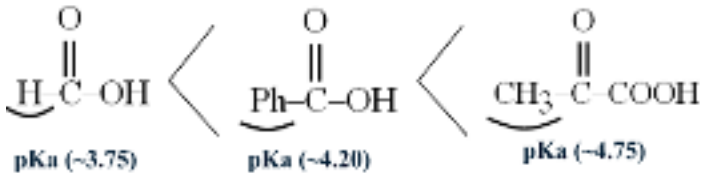
Theoretical

65)

The enol form is more stable than the keto form if it is stabilized by intramolecular hydrogen bonding and resonance. A beta-diketone has a high enol content.
Enol form is aromatic in nature



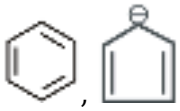
Final Answer : Option (1)

66) 

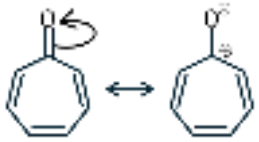
Acidic strength : x > z > y
Benzoic acid has acidic strength in between formic acid and acetic acid.
Final Answer: Option (1)

67)

An aromatic compound has complete cyclic resonance of (4n+2) π electrons and is planer, (n = 0, 1, 2, ...)

 both have 6πe⁻ in complete cyclic resonance.

 ⇒ 2πe⁻ in complete cyclic resonance.

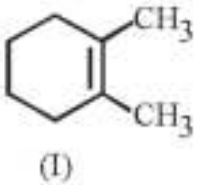
 ⇒ 6πe⁻ in complete cyclic resonance.

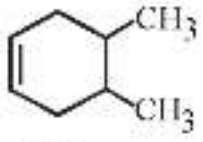
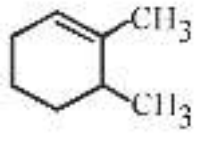
Total no. of aromatic compounds are 4.
Final Answer : Option (2)

68)

Concept: Heat of HYDROGENATION [HOH] $\propto \frac{1}{\text{stability of Alkene}}$
→ stability of Alkene ∝ Resonance ∝ Number of α- H.

Solution:

Given compounds	**Remark**
 (I)	**α - H ⇒ 10**

 (II)	**α - H ⇒ 4**
 (III)	**α - H ⇒ 6**

 stability of Alkene :- (I) > (III) > (II)
order of HOH is :- (II) > (III) > (I)
Final answer: 3

69)

Theoretical

70)

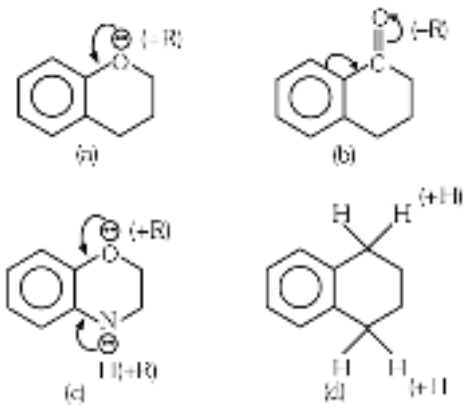
Theoretical

71)

Theoretical

72)

Reactivity towards electrophilic aromatic substitution reaction (ESR) ∝ density in ring.



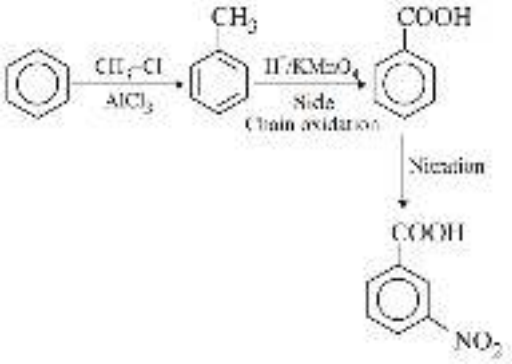
(2) c > a > d > b
Final answer : Option(2)

73) Friedel crafts acylation.

74)

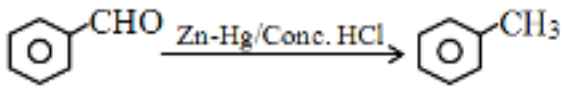
Theoretical

75)



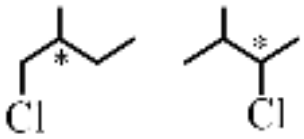
Benzoic acid is meta directing.

Final answer : Option (1)

76) 

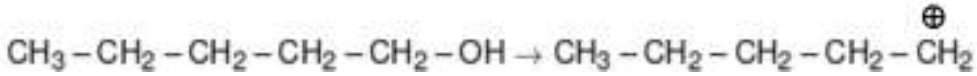
77)

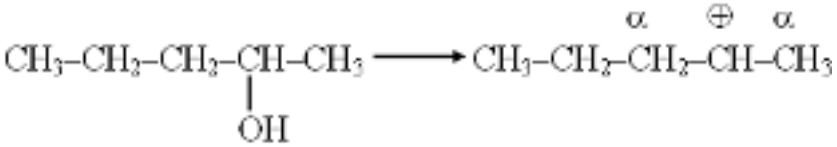
Theoretical

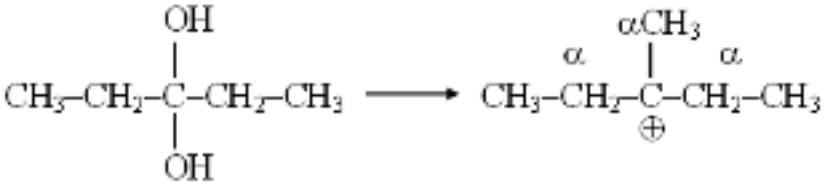
78) 

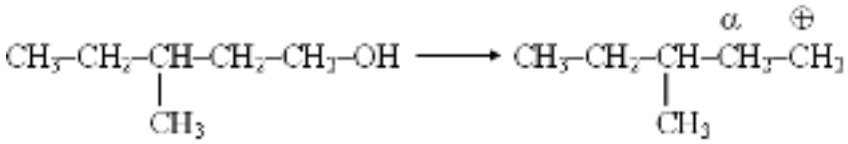
79)

Dehydration of alcohol ∝ stability of carbocation

(1)  αH = 2

(2)  αH = 5

(3)  αH = 7 most stable cation



(4) αH = 2

Option 3 tertiary alcohol forms the most stable carbocation intermediate. It will undergo dehydration most easily.

Final answer : option (3)

80)

Theoretical

81) NCERT 11th Page No. 389

82)

Theoretical

83)

Theoretical

84)

Electromeric effect is observed in compound having π bond.

85) NCERT Pg. # 386, 387

86)

Theoretical

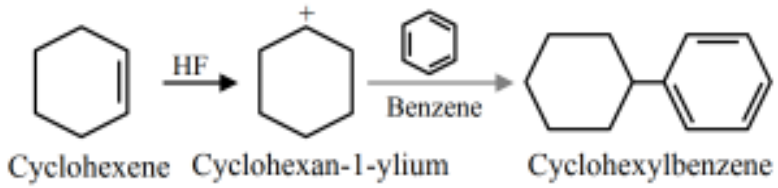
87)

Theoretical

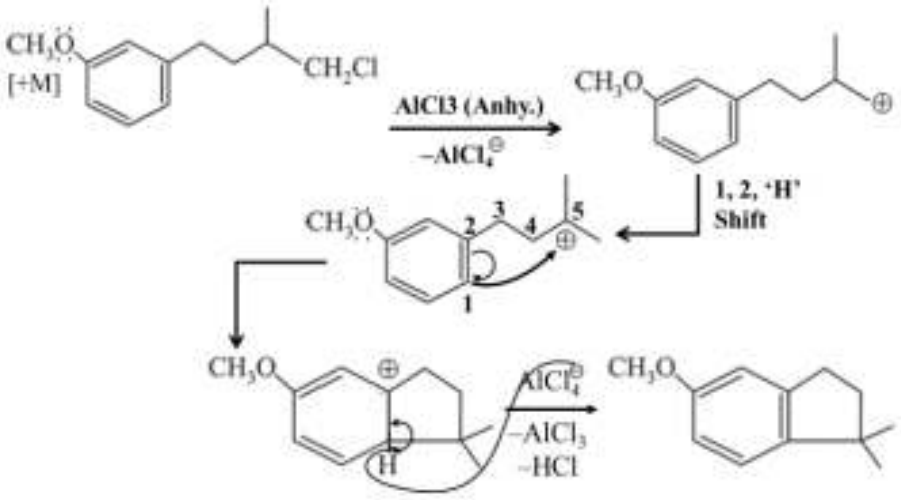
88)

Theoretical

89)



90)



BIOLOGY

91) NCERT Pg.# 134

92) NCERT, Pg. # 135

93) NCERT Pg.# 211

94) NCERT Pg. No. # 142

95) NCERT-XI, Pg # 218

96) NCERT-XI, Pg # 216

97) NCERT Pg. No. # 135

98) NCERT-XI, Pg # 223

99)

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100) NCERT Pg. # 135

101) NCERT Pg. # 213

102) NCERT Pg. # 143

103) NCERT Pg. # 220

104) NCERT Pg. No. # 147

105)

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106) NCERT Pg.#134,135

107) NCERT Pg. # 147

108)

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109) NCERT Pg. # 158

110) NCERT Pg. # 159

111) NCERT Pg. # 159,160

112) NCERT Pg. # 160

113) NCERT Pg # 163-164

114) NCERT, Pg. # 157

115) NCERT Pg. # 163

116)

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117) NCERT Pg. # 158

118) NCERT Pg. # 159

119) NCERT-XI Pg#160

120) NCERT Pg. # 160

121) NCERT, Pg # 163,164

122) NCERT, Pg. # 157

123) NCERT Pg#162

124)

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125) NCERT Pg. # 241

126) NCERT-XI, Pg # 248

127) NCERT XI Pg. # 247

128) NCERT-XI, Pg. # 246

129) NCERT-XI, Pg. # 248

130)

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131) NCERT-XI, Pg # 249

132) NCERT Pg. # 250

133) NCERT-XII, Pg. # 175

134)

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135)

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138) NCERT-XI, Pg. # 234

139)

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140) NCERT-XI, Pg. # 236-237

141) NCERT-XI Pg. # 237, 230

142) NCERT-XI, Pg. # 236

143)

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144) NCERT-XI, Pg # 232 (18.3)

145) NCERT-XI, Pg. # 233

146)

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148) NCERT-XI, Pg. # 248

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154) NCERT-XI Pg. # 249, 250, 251

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156) NCERT-XI, Pg. # 250

157) NCERT-XI, Pg. # 255

158) NCERT-XI, Pg. # 254

159) NCERT-XI, Pg. # 254

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161)

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162) NCERT-XI, Pg. # 252

163) NCERT-XI, Pg. # 250

164) NCERT-XI, Pg. # 212

165) NCERT-XI, Pg. # 255

166) NCERT-XI, Pg. # 248

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171)

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172) NCERT-XI, Pg. # 226, 227

173)

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174)

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175)

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176) NCERT-XI, Pg. # 222

177)

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178) NCERT-XI, Pg. # 227

179)

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